crucially, the equations show that this energy has a uniform, negative pressure. What's more, the gravitational repulsion of the cosmological constant's negative pressure overwhelms the gravitational attraction coming from its positive energy, and so repulsive gravity wins the competition: *a cosmo*logical constant exerts an overall repulsive gravitational force.[7]

For Einstein, this was just what the doctor ordered. Ordinary matter and radiation, spread throughout the universe, exert an attractive gravitational force, causing every region of space to *pull* on every other. The new cosmological term, which he envisioned as also being spread uniformly throughout the universe, exerts a repulsive gravitational force, causing every region of space to *push* on every other. By carefully choosing the size of the new term, Einstein found that he could precisely balance the usual attractive gravitational force with the newly discovered repulsive gravitational force, and produce a static universe.

Moreover, because the new repulsive gravitational force arises from the energy and pressure in space itself, Einstein found that its strength is cumulative; the force becomes stronger over larger spatial separations, since more intervening space means more outward pushing. On the distance scales of the earth or the entire solar system, Einstein showed that the new repulsive gravitational force is immeasurably tiny. It becomes important only over vastly larger cosmological expanses, thereby preserving all the successes of both Newton's theory and his own general relativity when they are applied closer to home. In short, Einstein found he could have his cake and eat it too: he could maintain all the appealing, experimentally confirmed features of general relativity while basking in the eternal serenity of an unchanging cosmos, one that was neither expanding nor contracting.

With this result, Einstein no doubt breathed a sigh of relief. How heart-wrenching it would have been if the decade of grueling research he had devoted to formulating general relativity resulted in a theory that was incompatible with the static universe apparent to anyone who gazed up at the night sky. But, as we have seen, a dozen years later the story took a sharp turn. In 1929, Hubble showed that cursory skyward gazes can be misleading. His systematic observations revealed that the universe is *not* static. It *is* expanding. Had Einstein trusted the original equations